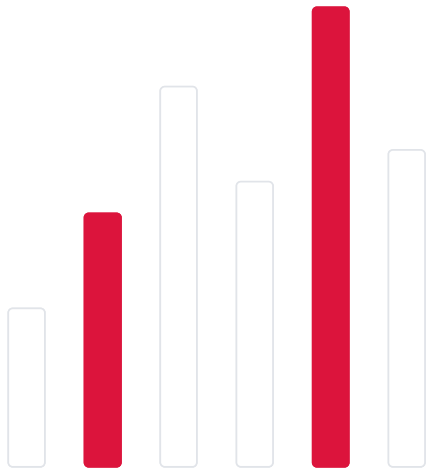
POWER BI PROJECT PRESENTATION

Airtel Telecom Analytics Dashboard

Customer • Recharge • Revenue • Plan Insights

A data-driven view of customer behaviour and telecom performance



Presented by:
Vismaya N

01 / INTRODUCTION

Project Overview

This project uses Power BI to transform telecom customer and monthly recharge data into an interactive analytical report.

BUSINESS AREA

Telecommunications

Customer and plan performance

TOOL

Microsoft Power BI

Data modelling, DAX and visuals

REPORT PAGES

3

Overview • Customer • Plan

PROJECT PURPOSE

- Monitor revenue and recharge activity.
- Analyze customer satisfaction, complaints and churn.
- Compare plan and state-level performance.

Dataset Note: This project uses synthetic telecom data for demonstration and does not represent actual Airtel operational data.

Problem Statement & Objectives

THE CHALLENGE

Telecom data can be spread across customer , recharge , plans and locations.

A unified report makes performance easier to compare and patterns easier to identify.

PROJECT OBJECTIVES

- Create a clear, interactive dashboard.
- Track revenue, recharges and data usage.
- Analyze customer satisfaction, complaints and churn.
- Compare plan and state performance.
- Enable filtering by time and category.

Dataset & Data Preparation

The practice dataset contains synthetic telecom records for customer and monthly activity analysis.

CUSTOMER DATA 1,200	FACT TABLE Monthly_revenue	PLAN CATEGORIES 3
TOTAL TABLES 6	MONTHLY_REVENUE ROWS 20,100	TOTAL COLUMNS 9

MAIN FIELDS USED

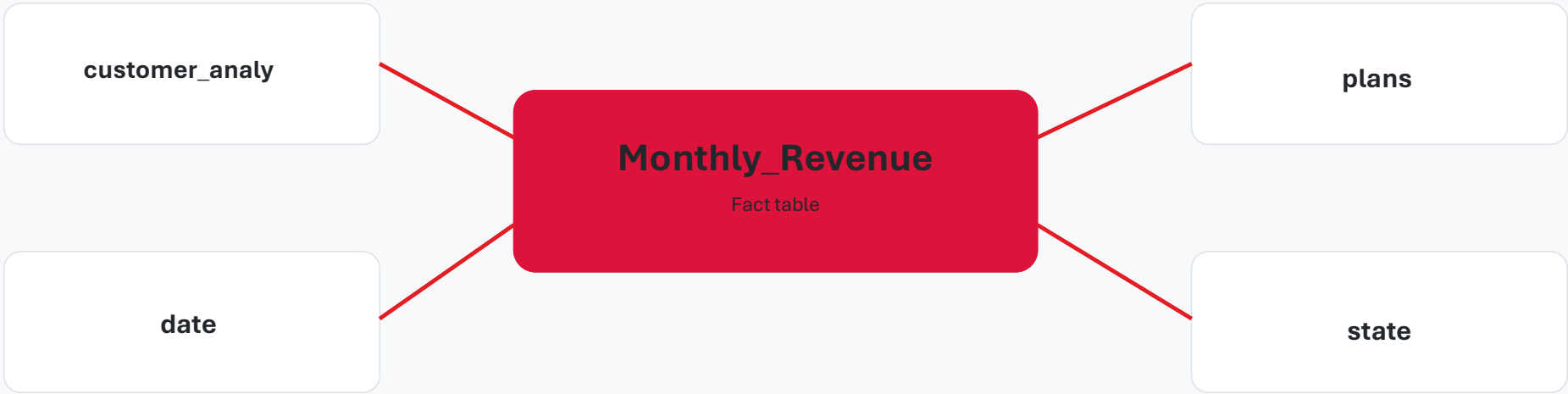
- **Customer** : Customer ID, join date, state, city and customer segment
- **Plan** & **Revenue** : Plan name/type, monthly price and revenue
- **Usage** & **Service** : Recharge count, data usage and satisfaction score Complaint status and churn flag

DATA CLEANING &TRANSFORMATION

Missing Values Checked and handled where necessary	Duplicates Checked for duplicate records and key values.	Data Types Verified date, numeric, and text columns	Power Query Used to prepare and transform data.

Data Model & Relationships

A relationship based data model connects monthly telecom activity with customer, date, plan, and state information.



Model structure: Monthly_Revenue act as the central fact table, connected to customer , date , plan and state table through relationship defined in Power BI . the relevant dimension tables.

Key Metrics & DAX Measures

Measures calculate key performance indicators across the report.

Total Revenue	SUM(Revenue_INR)
Total Recharges	SUM(Recharge_Count)
Monthly Customers	DISTINCTCOUNT(Customer_ID)
Average Satisfaction	AVERAGE(Satisfaction_Score)
Churn Rate	Churned customers ÷ total customers
Average Plan Price	AVERAGE(Monthly_Plan_Price_INR)

Validation: Measures were checked for accurate results and interactive filter behaviour .

Dashboard Page 1 — Overview

The Overview page provides a high-level snapshot of telecom performance through key KPIs and visualizations.

REVENUE	CUSTOMERS	RECHARGES	DATA USAGE	AVG REVENUE
₹11.11M	1,200	43,058	520K GB	₹ 9,259
TOTAL REVENUE	TOTAL CUSTOMERS	TOTAL RECHARGES	TOTAL DATA USAGE	AVG REVENUE PER CUSTOMER

MAIN INSIGHTS

- Monthly revenue and recharge trends
- Revenue by state and plan type
- Customer distribution and active-customer analysis
- Filters for date, state and plan performance

INTERACTIVITY

- Interactive slicers allow users to filter dashboard visuals by month , state , or plan type .
- Reset Filters button restores the default filter selections .

Dashboard Page 2 — Customer Analytics

This page explores customer satisfaction, complaints, segments and churn behaviour to understand customer experience and retention .

CUSTOMERS	SATISFACTION	CHURN RATE	COMPLAINTS	PLAN PRICE
1,200	3.51 / 5	9.6%	214	₹ 546
Total Customers	Average score	Customer Churn Rate	Customers with complaints	Average Plan Price

KEY INSIGHTS

- **Customer Satisfaction:** The average score of 3.51/5 indicates scope to improve customer experience.
- **Customer Retention:** The 9.6% churn rate highlights the importance of monitoring customer retention.
- **Customer Complaints:** The 214 customers with complaints highlight an opportunity to investigate customer issues and improve service quality.

Dashboard Page 3 — Plan & Revenue Analysis

This page compares plan types, plan prices, recharge activity and state-level revenue to evaluate telecom performance.

TOTAL REVENUE
₹11.11M
Revenue in dataset

RECHARGES
43,058
Total recharge events

AVG PLAN PRICE
₹546
Average monthly price

PLAN TYPES
3
Prepaid • Postpaid • Broadband

DATA USAGE
520 K GB
Total data usage

KEY COMPARISONS

- Top states by revenue
- Top plans by revenue
- Average plan price by plan type
- Revenue contribution and total recharges by plan type

INTERACTIVITY

Drill-through to the Plan Details page for more detailed plan-level analysis.

09 / WRAP-UP

Key Findings & Conclusion

KEY TAKEAWAYS

- Prepaid contributes the largest revenue share in the sample dataset.
- Average customer satisfaction is around 3.51 out of 5.
- Churn and complaints can be explored by segment, state and plan.
- Interactive Slicers and drill through support deeper data exploration .

LIMITATIONS & FUTURE ENHANCEMENT

- The analysis uses a synthetic dataset for demonstration purposes.
- Future enhancements could include revenue forecasting and customer retention analysis.
- Additional plan-level analysis could provide deeper business insights.
- Adding more interactive visuals could help users explore customer and revenue trends in greater detail.

CONCLUSION

The Power BI dashboard brings customer, revenue, recharge, and plan data together to support interactive analysis and data-driven decision-making.

THANK YOU

Airtel Telecom Analytics Dashboard

Interactive Data Analysis Using Power BI